SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Industry Problem Solving Task

COURSE NAME: R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO2: R Programming Structures, Different types of loops such as: for (), while() loops, Control Statements, Looping Over Non-Vector Sets, - If-Else, Arithmetic and Boolean Operators and values, Default Values for Argument, Return Values, Deciding Whether to explicitly call return- Returning Complex Objects, Functions are Objective, No Pointers in R, Recursion. (BL4-BL5)

Assessment Tool Weightage: 25%

Code of Conduct:

I Nikhil Kumar Reddy Kadiveti (192425031) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

QUESTIONS: **Total Marks: 100**

Scenario

A retail company operates stores across multiple cities. The management has collected customer transaction data containing:

- Customer_ID
- Gender
- Age
- City
- Product_Category
- Purchase_Amount
- Monthly_Income
- Customer_Rating
- Membership_Status
- Store_Type

The company wants to improve customer satisfaction, optimize marketing strategies, and increase sales using data-driven decisions.

Question 1

The management wants to understand the overall spending behavior of customers. Calculate and interpret appropriate measures of central tendency (mean, median, and mode) for Purchase Amount. Based on the results, recommend which measure best represents customer spending patterns.

Question 2

Analyze the variability in customer spending using measures of dispersion such as range, variance, standard deviation, and interquartile range. Identify whether spending behavior is consistent across customers and justify your findings.

Question 3

The marketing team suspects that customer ratings differ significantly between premium and regular members. Perform an exploratory analysis of Customer Rating and summarize the key patterns using suitable statistical measures and visualizations.

Question 4

Investigate the relationship between Monthly Income and Purchase Amount using bivariate analysis techniques. Interpret the strength and direction of the relationship and discuss its implications for targeted marketing campaigns.

Question 5

The company wants to determine whether male and female customers differ in their average purchase amount. Formulate appropriate hypotheses, perform a T-test, and interpret the results to support business decision-making.

Question 6

Management wishes to compare customer satisfaction ratings across three different store types (Urban, Semi-Urban, and Rural). Conduct an ANOVA test and explain whether significant differences exist among the store categories.

Question 7

Analyze whether Membership Status and Product Category preference are associated. Perform a Chi-square test, interpret the findings, and explain how the results can support customer retention strategies.

Question 8

The company plans to launch a new loyalty program. Using EDA techniques, identify the customer segment that should be targeted first. Support your recommendation using appropriate statistical evidence.

Question 9

Based on customer transaction data, identify key factors influencing high purchase amounts. Utilize central tendency measures, dispersion measures, and bivariate analysis to justify your conclusions.

Question 10

Prepare a business analytics report summarizing the insights obtained from EDA, T-test, ANOVA, and Chi-square analysis. Provide actionable recommendations that can help management improve customer engagement and profitability.

RUBRICS FOR CALCULATION

Industry Problem Solving Task					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Application of Techniques	25	Correctly applies central tendency, dispersion, and bivariate analysis techniques.	Applies most EDA techniques correctly with minor omissions.	Applies basic EDA techniques with limited accuracy.	Demonstrates limited or incorrect application of EDA techniques.
Statistical Analysis & Testing	25	Correctly performs and interprets T-test, ANOVA, and Chi-square analysis.	Performs statistical tests with minor interpretation errors.	Performs basic statistical analysis with limited interpretation.	Demonstrates poor understanding of statistical tests.
Problem Analysis & Data Interpretation	20	Accurately analyzes data patterns and derives meaningful insights.	Provides reasonable analysis with adequate interpretation.	Provides limited analysis and interpretation.	Analysis is weak or incorrect.
Decision-Making & Business Recommendations	20	Provides data-driven recommendations that effectively address the business problem.	Recommendations are relevant and supported by analysis.	Recommendations are basic with limited justification.	Recommendations are weak, unsupported, or missing.
Organization & Completeness	10	Response is well-structured, complete, and addresses all requirements.	Mostly complete and organized.	Partially complete with some missing components.	Incomplete and poorly organized.

Question 1 – Central Tendency of Purchase Amount

Table 1. Measures of central tendency (n = 500)

Measure	Value	Reading
Mean	₹2,126.65	Pulled upward by a few very large purchases
Median	₹1,613	Half of customers spend less than this
Mode (₹500 bin)	₹1,000 – ₹1,500	Most frequent spending band
10% trimmed mean	₹1,850	Mean after dropping the top/bottom 10%
Skewness	2.28	Strong positive (right) skew

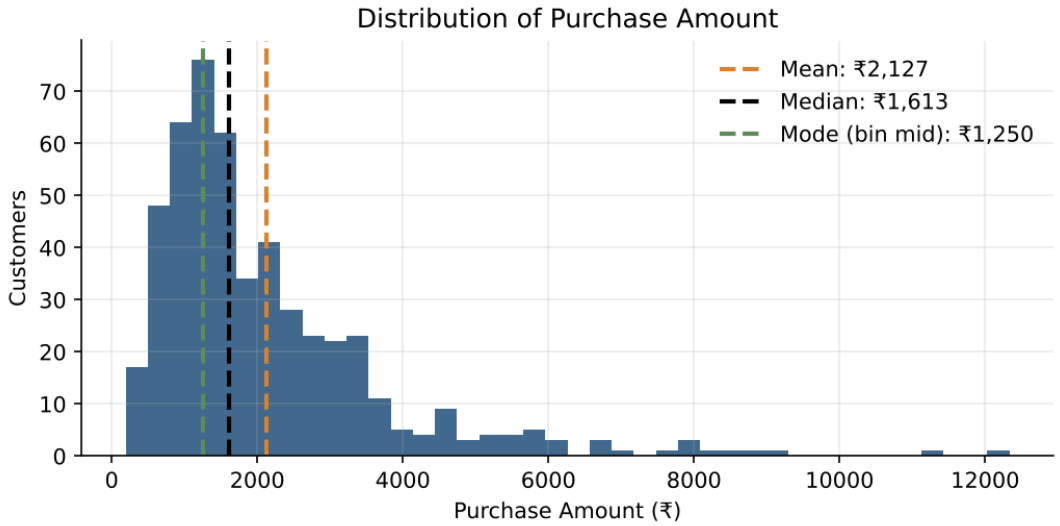


Figure 1. Purchase Amount with mean, median and mode marked.

Interpretation. Purchase Amount is a continuous variable, so the mode is taken as the most populated ₹500 bin (₹1,000–₹1,500). The ordering mode < median < mean (₹1,500 < ₹1,613 < ₹2,127) and a skewness of 2.28 show a right-skewed distribution: most customers make modest purchases while a minority of big-ticket buyers (mostly Electronics) stretch the tail and inflate the mean by about ₹514 (32% above the median).

Conclusion: The **median (₹1,613)** is the best single measure of typical customer spending because it is resistant to extreme values. The mean should be used only for revenue forecasting (mean × customers = total sales), and the modal band tells merchandising which price points are most common.

Question 2 – Variability in Customer Spending

Table 2. Measures of dispersion

Measure	Value	Note
Range	₹12,137	₹194 to ₹12,331
Variance	2,622,593 (₹^{2})	Sample variance (n − 1)
Standard deviation	₹1,619.44	Typical distance from the mean
Interquartile range	₹1,609	Q1 = ₹1,090.5, Q3 = ₹2,699.5
Coefficient of variation	76.1%	SD ÷ mean
Outliers (1.5 × IQR rule)	30 customers	Above ₹5,113

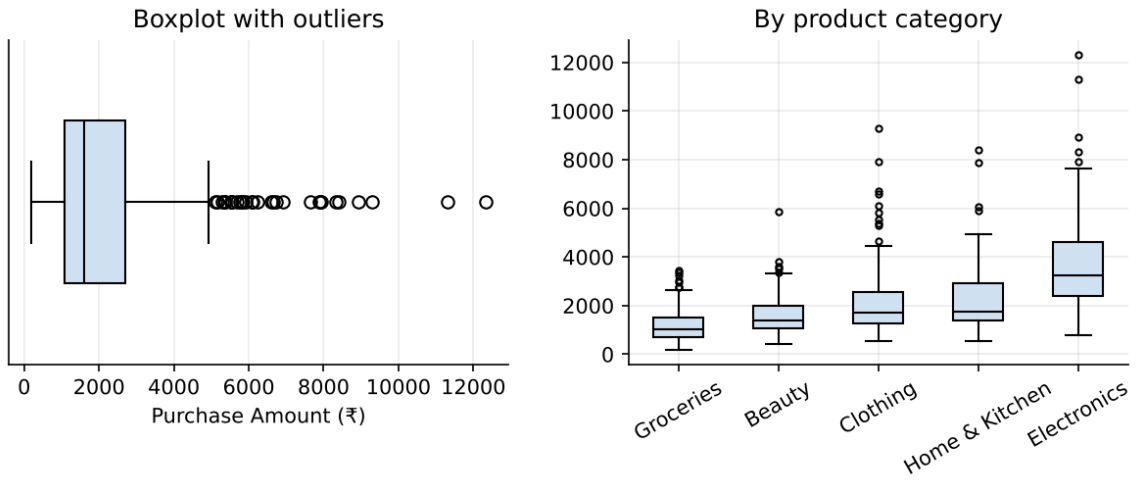


Figure 2. Boxplot of Purchase Amount (left) and by product category (right).

Table 3. Spending variability by product category

Category	Mean	Median	SD	CV
Electronics	₹3,728	₹3,244	₹2,107	56%
Home & Kitchen	₹2,292	₹1,765	₹1,516	66%
Clothing	₹2,204	₹1,728	₹1,487	67%
Beauty	₹1,666	₹1,400	₹932	56%
Groceries	₹1,215	₹1,027	₹685	56%

Interpretation. The standard deviation (₹1,619) is 76% of the mean – a CV above 50% is usually read as very high dispersion. The IQR (₹1,609) shows that even the middle 50% of customers span a wide band, and the range is more than 5.7 times the mean. The 30 outliers (6% of customers) all lie on the high side, which is why the SD and variance are inflated relative to the IQR. Variability is also uneven across categories: Electronics buyers spread widely (absolute SD ₹2,107) while Groceries spending is tight.

Conclusion: Spending behaviour is **not consistent** across customers. Customers are a mix of low-value routine shoppers and a small group of high-value buyers, so one average campaign will fit neither; the IQR (not the SD) is the more reliable description of the core customer.

Question 3 – Customer Rating: Premium vs Regular Members

Table 4. Customer Rating summary by membership

Group	n	Mean	Median	SD	IQR range
Premium	216	3.96	4.0	0.71	3.5 – 4.5
Regular	284	3.60	3.6	0.72	3.0 – 4.1
All customers	500	3.76	3.8	0.73	3.2 – 4.3

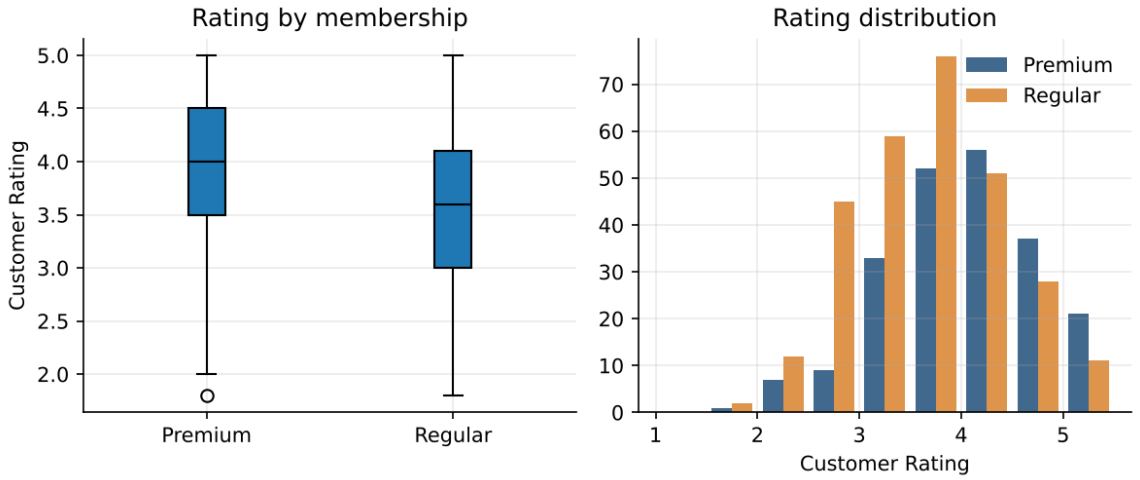


Figure 3. Customer Rating by membership: boxplot and histogram.

Interpretation. Premium customers rate the store about 0.36 points higher (mean 3.96 vs 3.60; median 4.0 vs 3.6). The histogram shows the whole Premium distribution shifted to the right, not just a few high scorers, and the spreads are similar (SD ≈ 0.7 in both). A Welch t-test confirms the difference is real: $t = 5.61$, $p < 0.001$; the Mann–Whitney test agrees ($p < 0.001$). Cohen's $d = 0.51$ is a **medium** effect, so the difference is practically meaningful as well as statistically significant.

Conclusion: The marketing team's suspicion is **supported**: Premium members are significantly more satisfied than Regular members. Note this is an association – we cannot tell from this data whether membership causes satisfaction or satisfied customers choose to upgrade.

Question 4 – Monthly Income vs Purchase Amount

Table 5. Bivariate statistics

Statistic	Value
Pearson correlation (r)	0.450 ($p < 0.001$)
Spearman rank correlation (ρ)	0.521 ($p < 0.001$)
Pearson r on log–log scale	0.518
Covariance	14,928,735
Regression slope	₹0.0356 per ₹1 of income (≈ ₹356 per extra ₹10,000)
R^2	0.203 (20% of variation explained)

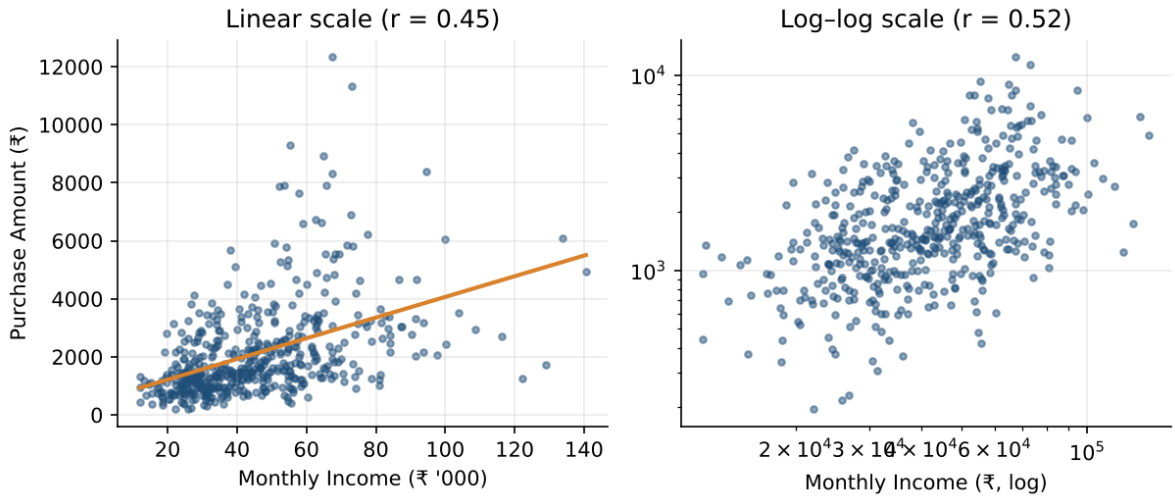


Figure 4. Income vs purchase: linear scale with fitted line (left) and log–log scale (right).

Interpretation. The relationship is **positive and moderate** (r = 0.45). Spearman's ρ (0.52) and the log–log correlation (0.52) are higher than Pearson's r because the relationship is monotonic but curved and heteroscedastic – the scatter fans out at high incomes, as the left panel shows. Each additional ₹10,000 of monthly income is associated with about ₹356 more per purchase on average, but income alone explains only 20% of the variation, so other factors (category, membership) matter too.

Conclusion: Income is a useful but incomplete targeting variable. Campaigns for **high-ticket categories** should be aimed at mid-to-high-income households, whereas low-income customers should receive value/basket-builder offers. Because the link is correlation, not proof of cause, income bands should be combined with category and membership when building audiences.

Question 5 – Do Male and Female Customers Spend Differently? (T-test)

Hypotheses. H_0: μ(Male) = μ(Female) (no difference in mean purchase amount). H_1: μ(Male) ≠ μ(Female) (two-sided). α = 0.05.

Table 6. Group statistics

Gender	n	Mean	SD
Male	271	₹2,188.43	₹1,718.87
Female	229	₹2,053.55	₹1,493.69

Table 7. T-test results and robustness checks

Test	Statistic	p-value	Decision
Levene (equal variances?)	F = 0.53	0.465	Variances comparable
Welch t-test (primary)	t(498) = 0.939	0.348	Fail to reject H_0
Pooled t-test	t = 0.928	0.354	Fail to reject H_0
t-test on log(purchase)	t = 0.956	0.340	Fail to reject H_0
Mann–Whitney U	–	0.526	Fail to reject H_0

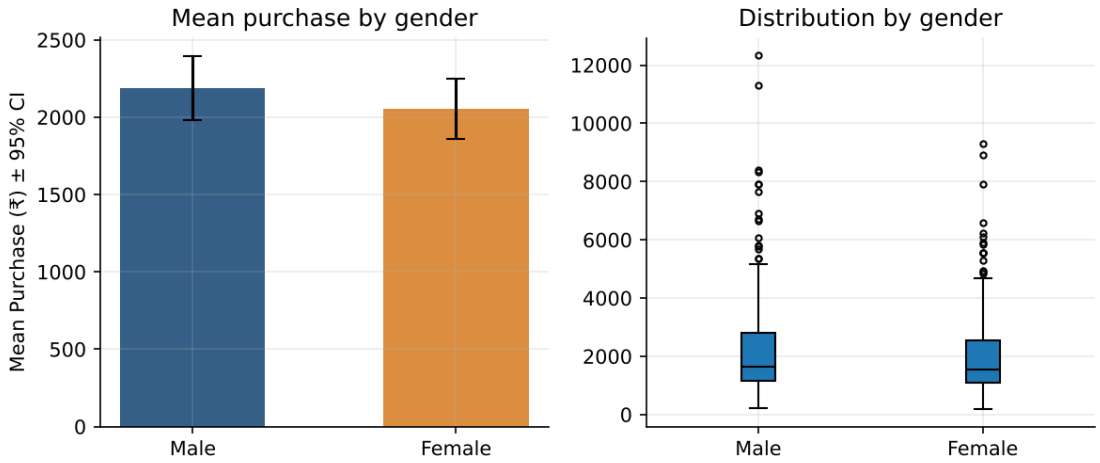


Figure 5. Mean purchase by gender with 95% confidence intervals (left) and distributions (right).

Interpretation. Men spent ₹135 more on average (₹2,188 vs ₹2,054), but the Welch t-test gives t(498) = 0.94, p = 0.348 – well above 0.05, and |t| is below the critical value of 1.96. The 95% confidence interval for the difference (₹-147 to ₹417) contains zero, and the effect size is negligible (Cohen's d = 0.08). The log-transformed test and the non-parametric test lead to the same decision, so the result does not depend on the skewness of the data.

Conclusion: We **fail to reject H_0**: there is no statistically significant difference in average purchase amount between men and women. Gender should not be used to segment pricing or campaigns on spend; targeting should rely on income, category and membership instead. (Failing to reject H_0 is not proof of equality; with n = 500 only differences larger than a few hundred rupees could have been detected reliably.)

Question 6 – Customer Satisfaction Across Store Types (ANOVA)

Hypotheses. H_0: μ(Urban) = μ(Semi-Urban) = μ(Rural). H_1: at least one store-type mean rating differs. α = 0.05.

Table 8. Rating by store type

Store type	n	Mean rating	SD
Urban	292	3.85	0.70
Semi-Urban	139	3.69	0.72
Rural	69	3.50	0.82

Table 9. One-way ANOVA table

Source	SS	df	MS	F	p-value
Between stores	7.89	2	3.943	7.53	0.0006
Within stores (error)	260.25	497	0.524		

Table 10. Tukey HSD post-hoc comparisons

Pair	Mean difference	Tukey adj. p	Significant?
Urban – Semi-Urban	0.16	0.0867	No
Urban – Rural	0.36	0.0008	Yes
Semi-Urban – Rural	0.20	0.1525	No

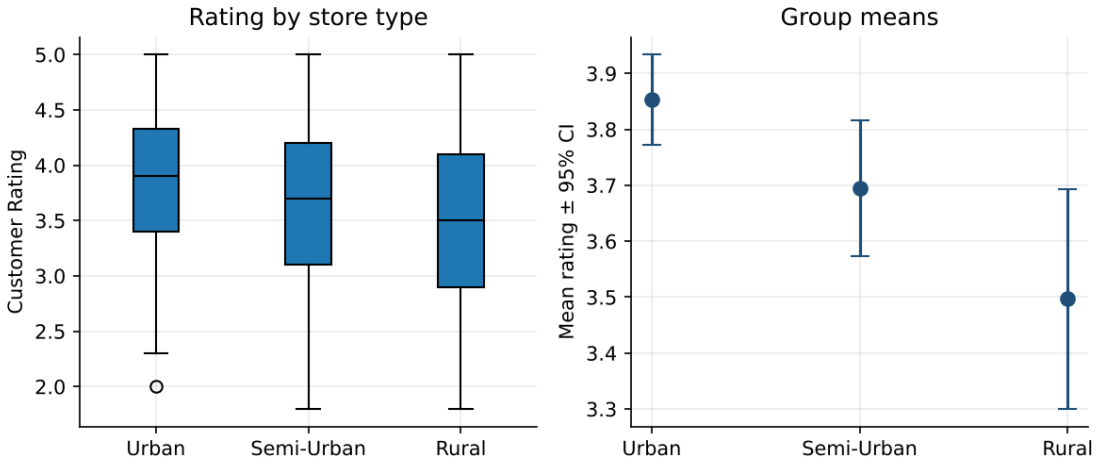


Figure 6. Rating by store type: boxplot and group means with 95% CI.

Interpretation. F(2, 497) = 7.53 exceeds the critical value 3.01, and $p < 0.001$, so H_0 is rejected. Levene's test (0.22) shows equal variances are reasonable and the Kruskal–Wallis test agrees (p = 0.002). Ratings fall from Urban (3.85) to Semi-Urban (3.69) to Rural (3.50). Tukey's test shows that the only pair that differs significantly is **Urban vs Rural** (difference 0.36, $p < 0.001$). The effect is small ($\eta^2 = 0.029$: store type explains about 3% of rating variance), and Rural also has the smallest sample (n = 69).

Conclusion: Significant differences **do exist**, driven by Rural stores scoring lower than Urban stores. Management should investigate service level, stock availability and delivery in Rural stores first; the Semi-Urban stores are statistically indistinguishable from either extreme.

Question 7 – Membership Status and Product Category (Chi-square)

Hypotheses. H_0: Membership Status and Product Category are independent. H_1: They are associated. $\alpha = 0.05$.

Table 11. Observed counts

Membership	Electronics	Clothing	Groceries	Home & Kitchen	Beauty	Total
Premium	46	65	31	36	38	216
Regular	40	64	108	38	34	284

Table 12. Expected counts under independence (all ≥ 5, so the test is valid)

Membership	Electronics	Clothing	Groceries	Home & Kitchen	Beauty
Premium	37.2	55.7	60.0	32.0	31.1
Regular	48.8	73.3	79.0	42.0	40.9

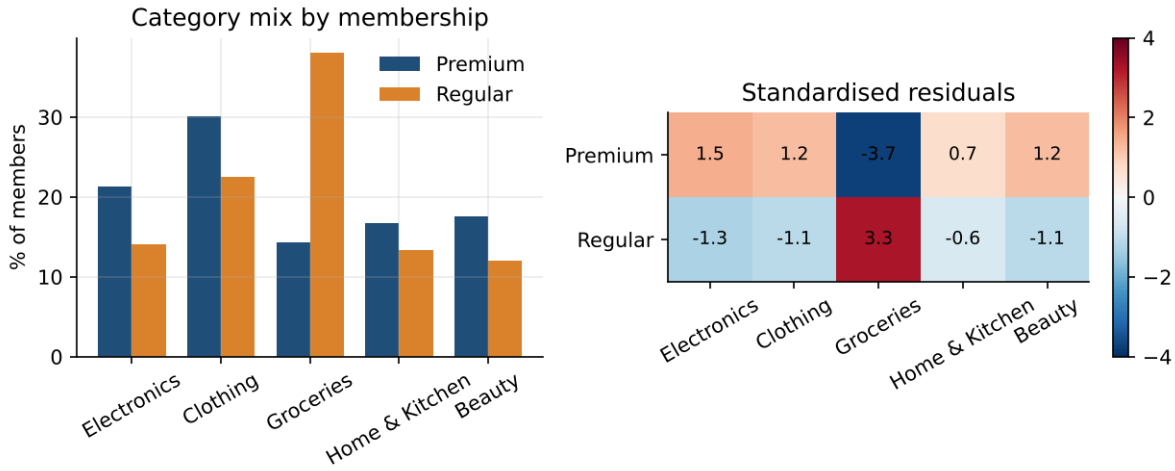


Figure 7. Category mix by membership (left) and standardised residuals (right).

Result. $\chi^2(4) = 34.75$ (critical value 9.49), $p < 0.001$; Cramér's V = 0.26, a moderate association. The smallest expected count is 31.1, so the Chi-square assumptions are met.

Interpretation. The pattern is concentrated in **Groceries**: 38% of Regular members' purchases are Groceries versus 14% for Premium members (standardised residuals +3.3 and -3.8). Premium members buy proportionally more Electronics (21% vs 14%) and Clothing (30% vs 22%). Residuals beyond ±2 mark the cells that drive the result; only Groceries exceeds that threshold.

Conclusion: H_0 is **rejected** – membership and category preference are related. For retention: (1) Regular members are anchored to low-ticket Groceries, so cross-sell Clothing and Electronics through bundles and first-purchase coupons to move them towards Premium; (2) protect Premium retention with early access and extended warranties in Electronics and Clothing, the categories they favour.

Question 8 – Segment to Target First for the Loyalty Programme

Customers were divided into income terciles (Low/Mid/High) and age bands (18–25, 26–35, 36–45, 46–65). For each segment the analysis measured size, average purchase, rating, share already Premium, and an **opportunity score** = total segment spend × share of customers not yet Premium (spend that a new loyalty programme could still influence).

Table 13. Segment ranking (all 12 segments)

Rank	Segment (income, age)	n	Avg purchase	Rating	% Premium	Opportunity (₹)
1	High, 36-45	56	₹3,343	3.70	57%	80,242
2	Mid, 36-45	59	₹2,088	3.80	41%	73,093
3	Low, 26-35	66	₹1,252	3.70	27%	60,101
4	High, 26-35	47	₹3,143	3.80	62%	56,575
5	Mid, 26-35	53	₹1,806	3.80	45%	52,362
6	Low, 36-45	42	₹1,447	3.60	24%	46,290
7	High, 46-65	42	₹2,836	3.80	64%	42,537
8	High, 18-25	22	₹2,659	3.80	41%	34,564
9	Mid, 46-65	24	₹2,492	3.80	50%	29,907
10	Low, 46-65	30	₹1,374	3.70	30%	28,854
11	Low, 18-25	29	₹1,339	3.90	28%	28,121
12	Mid, 18-25	30	₹1,620	3.80	47%	25,921

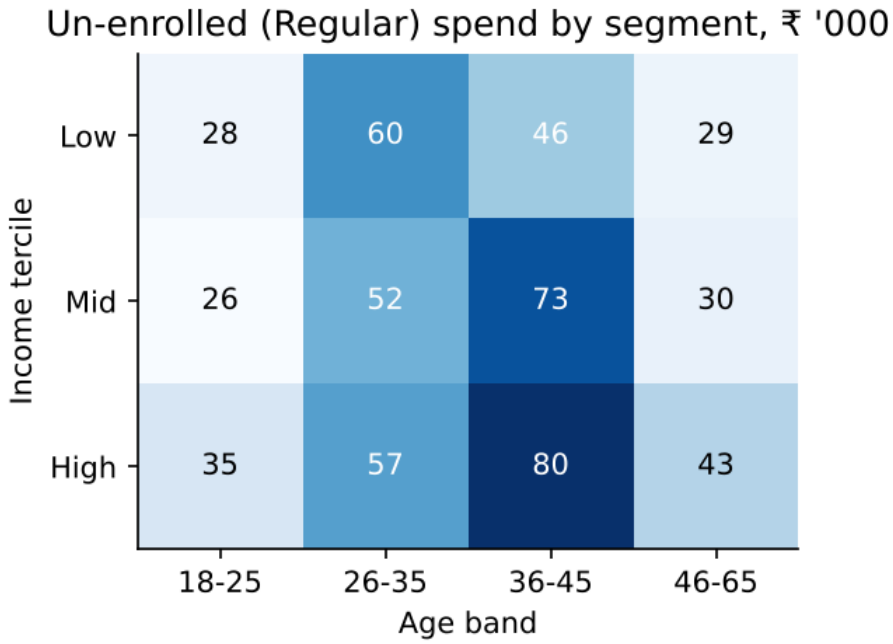


Figure 8. Spend by Regular (not Premium) customers in each segment, ₹ thousand.

Statistical evidence. (i) Premium customers spend significantly more per purchase than Regular customers (₹2,726 vs ₹1,670; Welch t = 7.16, p < 0.001). (ii) Membership depends on income (χ^2 = 33.2, p < 0.001): the share of Premium customers rises from 27% (Low) to 45% (Mid) to 58% (High). (iii) Mean spend rises with income band (₹1,338 → ₹1,972 → ₹3,069; Q4),

whereas average ratings are flat across segments (≈ 3.6–3.9), so spend and enrolment – not satisfaction – are what separate segments.

Recommendation: Target first: High-income customers aged 36–45 – the highest average purchase (₹3,343), the largest total spend (₹187,231) and the largest opportunity score (80,242). **Second wave:** Mid-income customers aged 36–45 (large, only 41% Premium). Caution: each segment contains only about 20–65 customers, so these rankings are directional and should be validated with a pilot.

Question 9 – Key Factors Behind High Purchase Amounts

A **high purchase** is defined as the top quartile of Purchase Amount (≥ ₹2,700, 125 customers). Central tendency, dispersion and bivariate analysis are used to find what distinguishes these customers.

Table 14. Numeric factors

Factor	High spenders (mean / median)	Others (mean / median)	r (point-biserial)	p (Mann–Whitney)
Monthly Income	59,242.4 / 58,800.0	41,043.5 / 37,800.0	0.39	< 0.001
Age	38.6 / 40.0	35.2 / 34.0	0.14	< 0.001
Customer Rating	3.8 / 3.9	3.7 / 3.8	0.05	0.347

Table 15. Categorical factors (baseline share of high spenders = 25%)

Factor	Level	n	% high spenders	Mean (₹)	Median (₹)	SD (₹)
Product Category	Electronics	86	67.4%	3,728	3,244	2,107
Product Category	Clothing	129	22.5%	2,204	1,728	1,487
Product Category	Groceries	139	5.0%	1,215	1,027	685
Product Category	Home & Kitchen	74	28.4%	2,292	1,765	1,516
Product Category	Beauty	72	13.9%	1,666	1,400	933
Membership Status	Premium	216	38.4%	2,726	2,182	1,916
Membership Status	Regular	284	14.8%	1,670	1,336	1,162
Gender	Male	271	26.9%	2,188	1,631	1,719
Gender	Female	229	22.7%	2,054	1,561	1,494
Store Type	Urban	292	24.3%	2,126	1,610	1,616
Store Type	Semi-Urban	139	28.8%	2,249	1,946	1,729
Store Type	Rural	69	20.3%	1,884	1,444	1,384

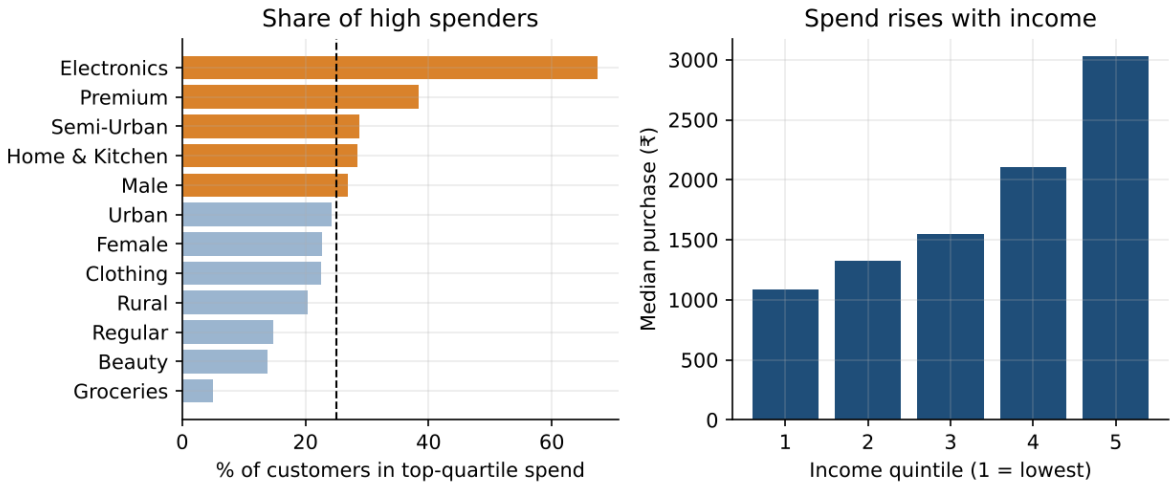


Figure 9. Share of top-quartile spenders by group (left) and median spend by income quintile (right).

Multivariable check. A regression of log(Purchase Amount) on all factors ($R^2 = 0.59$) isolates each factor's independent effect: a 1% rise in income is associated with a 0.68% rise in purchase amount ($p < 0.001$); Electronics purchases are about 103% higher than Beauty and Groceries about 28% lower; Regular members spend about 18% less than Premium members, other things equal. Age, gender and customer rating are not significant once income and category are controlled for (all $p > 0.05$).

Conclusion: The drivers of high purchase amounts, in order of strength, are **(1) Monthly Income** (high spenders' median ₹58,800 vs ₹37,800), **(2) Product Category – Electronics** (67% of Electronics buyers are top-quartile versus 5% in Groceries), and **(3) Premium membership** (38% vs 15%). Gender, store type and satisfaction rating do not distinguish high spenders. High spenders are also the most variable group, so averages should be supplemented by medians when reporting them.

Question 10 – Business Analytics Summary and Recommendations

10.1 Summary of statistical evidence

Table 16. Evidence summary

Analysis	Question	Result	Business meaning
EDA – central tendency	Typical spend?	Median ₹1,613 < mean ₹2,127; skew 2.3	Use the median to describe typical customers
EDA – dispersion	Consistent spend?	CV 76%; 30 high outliers	Customers are heterogeneous; segment campaigns
EDA – bivariate	Income vs spend?	r = 0.45; ρ = 0.52	Income is a key targeting variable
EDA – ratings	Premium vs Regular?	3.96 vs 3.60; d = 0.51	Premium customers are happier
Welch T-test	Gender and spend?	p = 0.348; not significant	No gender-based strategy
One-way ANOVA	Rating by store?	F = 7.53; p < 0.001; Urban > Rural	Fix Rural store experience
Chi-square	Membership × category?	χ^2 = 34.8; p < 0.001; V = 0.26	Regular members skew to Groceries; cross-sell

10.2 Actionable recommendations

1. **Launch the loyalty programme with High-income, 36–45 customers, then Mid-income, 36–45.** Highest opportunity score; Premium customers spend ₹1,056 more per purchase and are more satisfied (Q3, Q8). Run a pilot in this segment and compare against a control group before roll-out.
2. **Build tiered, income-aware offers instead of one blanket promotion.** Spending is highly dispersed (CV 76%) and rises with income (Q2, Q4). Use premium-product and early-access offers for high income; value bundles for low income.
3. **Use category cross-selling to move Regular members up.** Regular members are concentrated in Groceries; Premium members favour Electronics and Clothing (Q7). Offer Groceries-to-Clothing/Electronics bundles and 'upgrade to Premium' coupons at checkout.
4. **Prioritise Rural store service improvement.** Rural ratings are significantly below Urban (Q6). Audit staffing, stock availability and delivery; track the Rural rating monthly with a target of the Semi-Urban mean.
5. **Do not segment campaigns by gender.** No significant difference in spend (Q5); effort is better spent on income, category and membership targeting.
6. **Report spend with medians and quartiles, and monitor high-value outliers.** The mean overstates typical spend by ~32% (Q1). Create a VIP watch-list for customers above the outlier fence so the top buyers are retained personally.
7. **Grow Electronics sales to mid/high-income households.** Electronics buyers are most likely to be high spenders (Q9) and the highest-value category; feature financing/EMI and extended warranty.

10.3 Expected impact and KPIs

Initiative	KPI to monitor	Review cycle
Loyalty pilot (36–45, high/mid income)	Enrolment rate, repeat-purchase frequency, avg basket value vs control	Monthly
Cross-selling for Regular members	Share of non-Grocery purchases, Regular → Premium conversions	Monthly
Rural store improvement	Mean Customer Rating by store type, complaint rate	Monthly
High-value watch-list	12-month retention of customers above outlier fence	Quarterly

10.4 Limitations

- The analysis uses a synthetic dataset built to the specified structure; real-world effect sizes and rankings may differ. Re-run the Appendix script on the company data before acting.
- The tests show association, not causation (for example, Premium membership and satisfaction). Segment-level comparisons in Q8 rest on small cells ($n \approx 20$–65) and several tests were run at $\alpha = 0.05$ without multiple-testing correction, so borderline findings should be confirmed.
- Customer Rating is a bounded score (1–5); t-test and ANOVA are robust at this sample size, and non-parametric tests were reported alongside as a check.

Appendix – R Script (retail_eda.R)

The complete script below reproduces every statistic and chart in this report using base R only. It expects retail_transactions.csv in the working directory.

```
# ======================================================================
# ITA04 - R Programming | CO2-AT2 Industry Problem Solving Task
# Retail customer analytics: EDA, T-test, ANOVA, Chi-square
# Base R only (no extra packages). Place retail_transactions.csv in the
# working directory (or replace it with the company's real data).
# ======================================================================
retail <- read.csv("retail_transactions.csv", stringsAsFactors = FALSE)
retail$Gender            <- factor(retail$Gender)
retail$Membership_Status <- factor(retail$Membership_Status)
retail$Store_Type        <- factor(retail$Store_Type, levels = c("Urban", "Semi-Urban",
"Rural"))
retail$Product_Category  <- factor(retail$Product_Category)
str(retail); summary(retail)
dir.create("figs_R", showWarnings = FALSE)


# ---------- helper functions (default arguments, return values) -------
get_mode <- function(x, width = 500) {            # mode of a continuous variable (binned)
  bins <- floor(x / width) * width
  tab  <- table(bins)
  lo   <- as.numeric(names(tab)[which.max(tab)])
  return(c(lower = lo, upper = lo + width))
}
describe <- function(x, digits = 2) {              # returns a complex object (named list)
  q <- quantile(x, c(0.25, 0.75))                  # type 7, R default
  list(n = length(x), mean = round(mean(x), digits), median = median(x),
       sd = round(sd(x), digits), var = round(var(x), digits),
       range = diff(range(x)), IQR = unname(q[2] - q[1]),
       CV_pct = round(sd(x) / mean(x) * 100, digits))
}
interpret_p <- function(p, alpha = 0.05) {         # control statement (if / else)
  if (p < alpha) "Reject H0: statistically significant"
  else           "Fail to reject H0: not significant"
}
cohens_d <- function(a, b) {
  sp <- sqrt(((length(a) - 1) * var(a) + (length(b) - 1) * var(b)) / (length(a) + length(b) -
2))
  (mean(a) - mean(b)) / sp
}


# ---------------------------- Q1 -----------------------------------
x <- retail$Purchase_Amount
cat("Mean:", mean(x), " Median:", median(x), " Mode bin:", get_mode(x), "\n")
cat("Skewness:", mean((x - mean(x))^3) / mean((x - mean(x))^2)^1.5, "\n")
png("figs_R/q1_hist.png", 900, 500)
hist(x, breaks = 40, col = "steelblue", main = "Distribution of Purchase Amount", xlab =
"Purchase Amount")
abline(v = c(mean(x), median(x)), col = c("orange", "black"), lty = 2, lwd = 2)
legend("topright", c("Mean", "Median"), col = c("orange", "black"), lty = 2, lwd = 2);
dev.off()


# ---------------------------- Q2 -----------------------------------
print(describe(x))
iqr <- IQR(x); fences <- quantile(x, c(.25, .75)) + c(-1.5, 1.5) * iqr
cat("Outliers:", sum(x < fences[1] | x > fences[2]), "\n")
```

```
for (cat_name in levels(retail$Product_Category)) {            # for loop over a non-vector
set
  s <- x[retail$Product_Category == cat_name]
  cat(sprintf("%-15s mean=%8.1f  median=%7.1f  sd=%7.1f  CV=%5.1f%%\n",
              cat_name, mean(s), median(s), sd(s), sd(s) / mean(s) * 100))
}
png("figs_R/q2_box.png", 900, 500); boxplot(x ~ retail$Product_Category, col = "lightblue",
las = 2); dev.off()


# ---------------------------- Q3 -----------------------------------
print(tapply(retail$Customer_Rating, retail$Membership_Status, describe))
P <- retail$Customer_Rating[retail$Membership_Status == "Premium"]
R <- retail$Customer_Rating[retail$Membership_Status == "Regular"]
print(t.test(P, R)); print(wilcox.test(P, R)); cat("Cohen d:", cohens_d(P, R), "\n")
png("figs_R/q3_rating.png", 900, 500); boxplot(Customer_Rating ~ Membership_Status, data =
retail, col = c("steelblue", "orange")); dev.off()


# ---------------------------- Q4 -----------------------------------
print(cor.test(retail$Monthly_Income, x, method = "pearson"))
print(cor.test(retail$Monthly_Income, x, method = "spearman", exact = FALSE))
print(cor.test(log(retail$Monthly_Income), log(x)))
fit <- lm(Purchase_Amount ~ Monthly_Income, data = retail); print(summary(fit))
png("figs_R/q4_scatter.png", 900, 500)
plot(retail$Monthly_Income, x, pch = 19, col = rgb(0, .3, .6, .4), xlab = "Monthly Income",
ylab = "Purchase Amount")
abline(fit, col = "orange", lwd = 2); dev.off()


# ---------------------------- Q5 -----------------------------------
M <- x[retail$Gender == "Male"]; Fm <- x[retail$Gender == "Female"]
print(var.test(M, Fm))
res5 <- t.test(M, Fm, var.equal = FALSE, alternative = "two.sided", conf.level = 0.95)
print(res5); cat(interpret_p(res5$p.value), "\n"); cat("Cohen d:", cohens_d(M, Fm), "\n")
print(t.test(log(M), log(Fm))); print(wilcox.test(M, Fm))


# ---------------------------- Q6 -----------------------------------
aov6 <- aov(Customer_Rating ~ Store_Type, data = retail)
print(summary(aov6)); print(TukeyHSD(aov6)); print(kruskal.test(Customer_Rating ~ Store_Type,
data = retail))
ss <- summary(aov6)[[1]]; cat("eta-squared:", ss[1, "Sum Sq"] / sum(ss[, "Sum Sq"]), "\n")
print(bartlett.test(Customer_Rating ~ Store_Type, data = retail))
png("figs_R/q6_anova.png", 900, 500); boxplot(Customer_Rating ~ Store_Type, data = retail, col
= "lightgreen"); dev.off()


# ---------------------------- Q7 -----------------------------------
tab <- table(retail$Membership_Status, retail$Product_Category); print(tab)
chi <- chisq.test(tab); print(chi); print(round(chi$expected, 1)); print(round(chi$stdres, 2))
cat("Cramer's V:", sqrt(unname(chi$statistic) / (sum(tab) * (min(dim(tab)) - 1))), "\n")
print(round(prop.table(tab, 1) * 100, 1))


# ---------------------------- Q8 -----------------------------------
retail$Age_Band    <- cut(retail$Age, c(17, 25, 35, 45, 65), labels = c("18-25", "26-35", "36-
45", "46-65"))
retail$Income_Band <- cut(retail$Monthly_Income, quantile(retail$Monthly_Income, 0:3 / 3),
                          labels = c("Low", "Mid", "High"), include.lowest = TRUE)
seg <- aggregate(cbind(Purchase_Amount, Customer_Rating) ~ Income_Band + Age_Band, data =
retail, FUN = mean)
seg$n          <- as.vector(table(retail$Income_Band, retail$Age_Band))
```

```
seg$premium_pct <- aggregate(I(Membership_Status == "Premium") ~ Income_Band + Age_Band,
retail, mean)[, 3] * 100
seg$total_spend <- aggregate(Purchase_Amount ~ Income_Band + Age_Band, retail,
sum)$Purchase_Amount
seg$opportunity <- seg$total_spend * (1 - seg$premium_pct / 100)   # spend not yet in the
loyalty tier
print(seg[order(-seg$opportunity), ])
print(t.test(Purchase_Amount ~ Membership_Status, data = retail))
print(chisq.test(table(retail$Income_Band, retail$Membership_Status)))


# ---------------------------- Q9 -----------------------------------
thr <- quantile(x, .75); retail$High <- retail$Purchase_Amount >= thr
for (v in c("Monthly_Income", "Age", "Customer_Rating"))
  cat(v, ": high mean =", mean(retail[retail$High, v]), " others =", mean(retail[!retail$High,
v]),
      " r_pb =", cor(retail$High, retail[[v]]), "\n")
for (v in c("Product_Category", "Membership_Status", "Gender", "Store_Type"))
  print(round(tapply(retail$High, retail[[v]], mean) * 100, 1))
model <- lm(log(Purchase_Amount) ~ Gender + Product_Category + Membership_Status + Store_Type
+
              log(Monthly_Income) + Age + Customer_Rating, data = retail)
print(summary(model))
```